Variation of Parameters for a System

For $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{f}(t)$, the **fundamental matrix** $\boldsymbol{\Phi}$ has independent complementary solutions as its columns, so $\boldsymbol{\Phi}' = \boldsymbol{A}\boldsymbol{\Phi}$.

Replace the constant vector in $\boldsymbol{y}_c = \boldsymbol{\Phi}\boldsymbol{c}$ by $\boldsymbol{u}(t)$. Substituting $\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{u}$ gives

$$\boldsymbol{\Phi}(t)\boldsymbol{u}'(t) = \boldsymbol{f}(t) \qquad \Longrightarrow \qquad \boldsymbol{u}'(t) = \boldsymbol{\Phi}(t)^{-1}\boldsymbol{f}(t).$$

One particular solution is

$$\boxed{\boldsymbol{y}_p(t) = \boldsymbol{\Phi}(t)\int_{t_0}^{t} \boldsymbol{\Phi}(s)^{-1}\boldsymbol{f}(s)\,ds},$$

where t_0 is any point in the interval of the system.

Worked Example with Complementary Solutions t^2 and t^3

Worked problem

Use variation of parameters for a system to find a particular solution of

$$t^2y'' - 4ty' + 6y = t^2, \qquad t > 0.$$

For this Euler equation, substitute $y = t^r$ into the left-hand side.

$$t^2y'' - 4ty' + 6y = \big[r(r-1) - 4r + 6\big]t^r = (r-2)(r-3)t^r.$$

The roots 2 and 3 give $y_1 = t^2$ and $y_2 = t^3$.

The First-Order System

On $(0, \infty)$, divide by t^2 and isolate y''.

$$y'' = \frac{4}{t}y' - \frac{6}{t^2}y + 1.$$

Set $\boldsymbol{z} = (y, y')^{\mathsf{T}}$. Its first component has derivative y' and its second has derivative y''.

$$\boldsymbol{z}' = \boldsymbol{A}(t)\boldsymbol{z} + \boldsymbol{f}(t), \qquad \boldsymbol{A}(t) = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix}, \qquad \boldsymbol{f}(t) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The Fundamental Matrix

Each complementary solution y_i gives a system solution $\boldsymbol{z}_i = (y_i, y_i')^{\mathsf{T}}$. Use those solutions as columns.

$$\boldsymbol{\Phi}(t) = \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = \begin{pmatrix} t^2 & t^3 \\ 2t & 3t^2 \end{pmatrix}.$$

Its determinant is the Wronskian of y_1 and y_2.

$$\det \boldsymbol{\Phi}(t) = 3t^4 - 2t^4 = t^4 > 0 \qquad \text{for } t > 0.$$

Thus the columns are independent and $\boldsymbol{\Phi}(t)$ is invertible throughout $(0, \infty)$.

The Varying Coefficients

Solve $\Phi(t)\boldsymbol{u}'(t) = \boldsymbol{f}(t)$.

$$\begin{pmatrix} t^2 & t^3 \\ 2t & 3t^2 \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The first row gives $u_1' = -tu_2'$. Substitution into the second gives $t^2 u_2' = 1$.

$$u_1' = -\frac{1}{t}, \qquad u_2' = \frac{1}{t^2} \qquad \Longrightarrow \qquad u_1 = -\log t, \quad u_2 = -\frac{1}{t}.$$

The integration constants multiply complementary solutions, so set them to zero to obtain a particular solution.

The Particular Solution and Its Check

The first component of $\boldsymbol{z}_p = \boldsymbol{\Phi u}$ gives the scalar solution.

$$t^2 u_1 + t^3 u_2 = -t^2 \log t - t^2.$$

Since $-t^2$ solves the homogeneous equation, we can drop it.

$$\boxed{y_p(t) = -t^2 \log t}$$

With $y_p' = -2t \log t - t$ and $y_p'' = -2 \log t - 3$, substitution gives

$$\begin{aligned} t^2 y_p'' - 4t y_p' + 6 y_p &= (-2 + 8 - 6) t^2 \log t + (-3 + 4) t^2 \\ &= t^2. \end{aligned}$$